

SHIVAM DOHARE

Frontend Engineer — React, Next.js, Performance & Real-Time Systems

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PROFESSIONAL SUMMARY

Frontend Engineer with **3+ years** specializing in performance-critical UIs and real-time data visualization across regulated domains. Built a **Cesium.js + React** satellite-tracking dashboard rendering **100+ live entities** at **sub-second WebSocket-driven latency**, and shipped **React / Next.js** medical-record workflows under strict bundle and **Lighthouse** budgets. Work closely with backend teams to shape APIs and data models for demanding frontend requirements.

TECHNICAL SKILLS

AI / Agentic: OpenAI SDK, RAG (retrieval-augmented generation), vector embeddings, Qdrant vector DB, MCP server authoring, tool-use / agent orchestration, multi-step agent loops, structured outputs, Zod output validation, prompt iteration and guardrails
Frontend: React.js, Next.js (App Router, SSR/SSG), JavaScript (ES6+), TypeScript, Redux, React Hook Form, Zod, Tailwind CSS
Backend: Node.js, Express.js, REST API design, WebSockets (socket.io / native ws), JWT auth, RBAC, API contract design
Database: MongoDB, Mongoose, Qdrant, schema design, indexing, aggregation pipelines, query optimization, state normalization patterns
Cloud & DevOps: AWS (EC2, S3, ECR/ECS), Docker, containerized deployments, CI-driven image builds
Styling & Testing: Material UI, Sass, Figma, design tokens, a11y · Jest, React Testing Library, Playwright, Postman, bundle-analyzer, Lighthouse
Specialized: Cesium.js (3D geospatial visualization), real-time data pipelines

SELECTED ACHIEVEMENTS

- ▶ Built a **Cesium.js + WebSocket** real-time satellite-tracking dashboard rendering **100+ live orbital entities** at **sub-second latency** — uncommon territory for frontend engineers.
- ▶ Reduced initial JS payload **~35%** and lifted **Lighthouse** performance from **~60 → 90+** on a production healthcare dashboard.
- ▶ Replaced HTTP polling with **WebSocket** subscriptions after profiling latency bottlenecks — cut refresh from **~5s** to **sub-second** on a real-time aerospace dashboard.
- ▶ Architected **JWT-based RBAC** for 4 user roles using **Next.js** middleware route guards and a typed permissions hook layer.

WORK EXPERIENCE

Minfy Technologies Jan 2025 – Present

Frontend Engineer · Healthcare — Medical Records Web Application (Jul 2025 – Present)

- ▶ Architected component structure for **5 modules** — patient records, lab reports, clinical workflows — in **React.js + Next.js**; defined component contracts with backend ahead of API delivery to unblock parallel work.
- ▶ Reduced initial JS payload **~35%** via **Next.js** route-level dynamic imports and **bundle-analyzer** audits that flagged unused dependencies.
- ▶ Improved **Lighthouse** performance from **~60 to 90+** by deferring non-critical scripts and migrating eager image loads to **next/image**.
- ▶ Architected **JWT-based RBAC** for 4 user roles using **Next.js** middleware route guards and a typed permissions hook for conditional UI rendering.
- ▶ Standardized form handling with **React Hook Form + Zod**, eliminating divergent validation logic across modules and aligning client rules with backend API contracts.

Frontend Engineer · Aerospace — Space Situational Awareness Dashboard (Jan – Jun 2025)

- ▶ Defined component architecture and data flow for a real-time satellite tracking dashboard in **Cesium.js + React.js**, rendering **100+ live orbital positions** and proximity alerts.
- ▶ Managed Cesium's imperative entity lifecycle from a **React subscription layer** to prevent memory leaks and stale entities across scene updates.
- ▶ Replaced 5-second HTTP polling with **WebSocket** subscriptions after profiling latency and redundant network load — cut dashboard refresh from **~5s** to **sub-second**.
- ▶ Authored a **useWebSocket** React hook handling reconnect logic; defined the event channels and message structure for live updates with the backend team.
- ▶ Built a reusable component system from **20+ Figma designs** — cards, panels, timeline scrubbers, alert toasts — backed by shared **design tokens**, eliminating UI duplication across operator workflows.

Kartikey Security Services

Aug 2024 – Dec 2024

Frontend Engineer | Defense Sector — Internal & Public Communication Portal

- ▶ Built **10+ user-facing features** in **React.js + Redux** for an internal communication portal — form-heavy workflows and data tables with render-optimized list views.
- ▶ Refactored Redux state to a **normalized entity-by-id** shape with **reselect-memoized selectors**, eliminating redundant re-renders.
- ▶ Added **Jest + React Testing Library** coverage for auth flow, forms, and data-table modules, reaching **75%+** on those surfaces.
- ▶ Resolved **30+ cross-browser issues** across Chrome, Firefox, Edge, and Safari — flexbox edge cases, date input handling, Safari IndexedDB quirks.
- ▶ Worked with the backend team to shape **Node.js / Express** endpoints where new response shapes were needed to support frontend workflows.

Kapston Services Limited

Apr 2023 – Jul 2024

Frontend Engineer | Defense Sector — Internal Communication Portal

- ▶ Owned the responsive UI for an internal portal serving **1,000+ daily active users**, standardizing layout patterns and shared components as the codebase grew.
- ▶ Cut page load time **~30%** via removing render-blocking scripts, system fonts on critical paths, and image compression + lazy loading.

PROJECTS

EscapeLab — E-Commerce UI with URL-Synchronized State & Performance Optimizations · View Live Demo

- ▶ Designed **URL-synchronized state architecture**: filters, sort, and pagination serialize to query params, enabling shareable filtered views and deterministic client-side state restoration.
- ▶ Engineered catalog rendering with **memoized cards**, **IntersectionObserver**-based image lazy-loading, and a **localStorage-persisted cart reducer** with optimistic updates.

Personal Portfolio Website · View Live Demo

- ▶ Built with **Next.js + Tailwind CSS**; used **next/image** for asset optimization, route prefetching for instant transitions, and metadata-driven SEO with **OpenGraph and JSON-LD**.
- ▶ Achieved **95+ Google Lighthouse** scores across Performance, Accessibility, Best Practices, and SEO categories.

EDUCATION

Bachelor of Engineering (B.E.) in Information Technology

2015 – 2019

University Institute of Engineering and Technology (UIET), Panjab University